

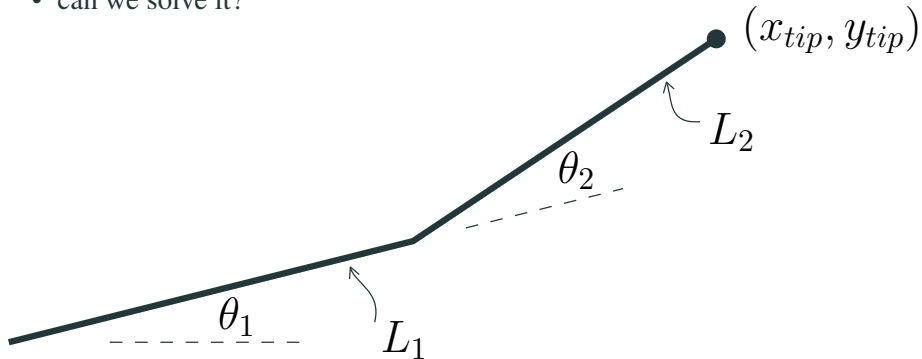
Two-Link Inverse Kinematics (Project 1)

EGR 445/545

GVSU

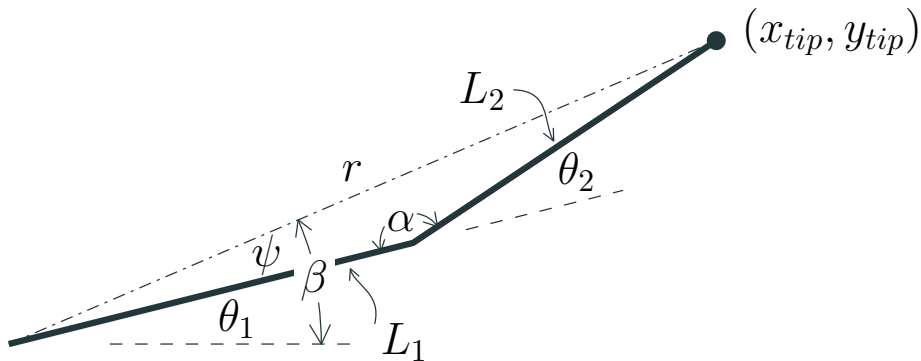
Two-Link Inverse Kinematics: Problem Statement

- what is known?
- what is unknown?
- can we write a relationship between x_{tip} , y_{tip} , θ_1 , and θ_2 ?
 - can we solve it?



Two-Link Inverse Kinematics: α , β , r , and ψ ?

- can we find α , β , r and ψ ?
- would finding them help?



Two-Link Inverse Kinematics: Document Camera

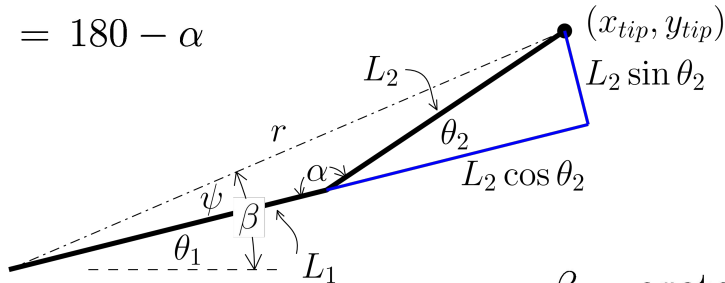
- let's do this...

Two-Link Inverse Kinematics Summary

$$r^2 = L_1^2 + L_2^2 - 2L_1L_2 \cos(\alpha)$$

$$\theta_2 = 180 - \alpha$$

$$r^2 = x_{tip}^2 + y_{tip}^2$$



$$\beta = \arctan2(y_{tip}, x_{tip})$$

$$\psi = \arctan2(L_2 \sin(\theta_2), L_1 + L_2 \cos(\theta_2)) \quad \theta_1 = \beta - \psi$$